

Article

Knowledge Graph Construction of End-of-Life Electric Vehicle Batteries for Robotic Disassembly

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Abstract: End-of-life (EoL) electric vehicle (EV) batteries are one of the main fountainheads for recycling rare metal elements like cobalt and lithium. Disassembly is the first step in carrying out a higher level of recycling and processing of EV batteries. This paper presents a knowledge graph of electric vehicle batteries for robotic disassembly. The information extraction of the EV batteries was conducted based on the source data of EV batteries. The semantic ontology structure and the knowledge graph of the EV batteries were constructed. A case study was designed to demonstrate the proposed knowledge graph. The study involved generating a robotic disassembly sequence planning for an EoL EV battery. The results show the feasibility of the constructed knowledge graph.

Keywords: robotic disassembly; electric vehicle battery; knowledge graph; disassembly sequence planning

1. Introduction

In 2012, Google Inc. originally proposed the concept of a knowledge graph in Google's Chrome browser search engine to provide users with better search experiences [1,2]. As structured semantic knowledge networks, knowledge graphs are gaining ever-increasing attention from industry, business, and academia, which offer relevant support for various scenarios, including intelligent search, technical support, and decision-making systems [3]. In addition, semantic web knowledge bases have been created and used, such as YAGO [4], FREEBASE [5], and DBpedia [6].

The recycling and reuse of EoL EV batteries have emerged as prominent approaches in recent years. The remanufacturing of EoL EV batteries has also gained recognition [7]. Human–robot collaborative disassembly and robotic disassembly of EoL EV batteries are more efficient and safer than traditional manual disassembly. The involvement of robots in disassembly operations can reduce time and save costs [8,9]. Human–robot collaborative disassembly has become a frontier by combining the advantages of robots and human operators [10].

An assembly information model based on a knowledge graph can facilitate the sharing of assembly process documentation information and increase the rate of information interaction during the development phase of the assembly process [11]. The advantage of EV battery knowledge graphs lies in the ability to display the first level, or even the second and third levels, of surrounding nodes intuitively on a graph. These surrounding nodes are connected to the starting node through relationships. EV battery knowledge graphs can provide detailed structural information to assist operators in understanding the layout and composition of batteries. In addition, planning the disassembly sequence using knowledge graphs can facilitate the robotic disassembly of EV batteries.

The structure of this paper is as follows. Section 2 outlines the current status of battery disassembly and the main problems in disassembly and recycling. Section 3 describes the construction process and results of using a knowledge graph for EV batteries. Section 4



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reports a case study of the disassembly sequence planning of EV batteries using the constructed knowledge graph. Section 5 concludes the paper.

2. Related Works

2.1. Processing Technical Levels of End-of-Life EV Batteries

Some scholars introduced the battery disassembly and recycling processes in terms of process improvements and technical levels. Pang Haifeng et al. [12] introduced the situation of lead–acid battery disassembly. The article pointed out that the disassembly and recycling technology level of lead–acid batteries is low, with high energy consumption and low metal recovery rates. Techniques referring to cleanliness, refinement, intelligence, and digitalization need to be improved. Disassembly equipment for the process of crushing and sorting EoL EV batteries was structured by Kang Fei et al. [13]. It improved the efficiency and digitalization of battery disassembly. Sonja Rosenberg et al. [14] explored how different disassembly steps affect EV battery disassembly time. The cost of a disassembly plant was also estimated. Due to the varying design architectures of EV batteries on the market, the condition and safety of these batteries during recycling remain uncertain.

The market is facing three major problems in EV battery disassembly and recycling [15]. Firstly, manufacturers produce batteries with various models and parameters. Secondly, in complicated disassembly tasks, the disassembly sequence becomes confusing, leading to low efficiency. Finally, manual disassembly has potential health and safety risks. Criteria were established for the EoL EV batteries and various options were highlighted for handling these batteries [16]. The concept of a second life for EV batteries was introduced and several key technologies related to EV batteries were identified to accelerate the large-scale industrialization of second-life batteries. Artificial intelligence and Big Data technologies showed promise in second-life batteries.

2.2. Robotic Disassembly of EV Batteries

Researchers proposed innovative equipment and automation methods for the complex processes of battery disassembly. Ren Wei et al. [17] introduced a robotic disassembly and motion planning system for EV batteries. Experiments were conducted on a robot simulation platform with and without obstacles. The experimental results demonstrated that the system was capable of independently planning and completing a task in dynamic environment. The utilization of telerobotic technology to explore a semi-robotic disassembly approach was proposed by Jamie Hathaway et al. [18]. These researchers assessed the success rate and completion time of telerobotic technology in tasks such as disassembling bolts, grasping, and removing cover plates. The results showed a significant reduction in the overall disassembly time. However, the effect of the different levels of expertise of operators was not considered.

A disassembly planning system was constructed by defining disassembly primitives and introducing neural predictions. The intelligent disassembly of EV batteries was employed by deploying robots [19]. The utilization of industrial robots for battery dismantling was explored [8]. The disassembly process was analyzed, and the operation was divided into clamping and cutting. Robotic disassembly of EV batteries improved security and saved time compared to manual disassembly. An information-driven robotic disassembly architecture was proposed by Hendrik Poschmann et al. [20]. This system included an information marketplace, robot cognition processor, system perception unit, disassembly execution unit, and human–machine interface. The system could incorporate information from the product’s entire lifecycle to ascertain the extent of disassembly.

2.3. Disassembly Sequence Planning and Knowledge Graph for the Disassembly of EV Batteries

Scholars created plans for disassembling returned parts that considered environmental factors and treatment methods. An automated optimization method was developed for planning disassembly sequences [21]. Components were classified in detail while separating them at the lowest level of disassembly to ensure the disposal of non-toxic, toxic, and

safe components. The management of parts after EoL treatment was proposed to reduce the environmental impact of the whole process [22]. A disassembly sequence was created by using the stability graph cut-set approach and setting the minimum number of direction changes as the fitness function. By considering the direction changes through the fitness function, the stability graph cut-set method was utilized to generate the optimal disassembly sequence. A scoring system was designed to process the waste by decomposing the components containing biohazardous toxic materials to the lowest level [23]. An environmental risk reduction model was investigated which included various parameters to tackle medical electronic waste. Artificial intelligence, knowledge engineering, deep learning, and mathematical algorithms were employed to analyze the disassembly sequence planning of EoL products with varying levels of scrap or disassembly requirements.

Artificial intelligence and machine learning were used to optimize the battery disassembly process. The approaches of employing artificial intelligence and machine learning to assist the disassembly of EV batteries were investigated by Kai Meng et al. [24]. A machine learning and sensor-based automatic disassembly platform for EoL batteries was demonstrated [25]. This platform combined a computer vision system and a thermal imager to enable real-time control of the cutting action and to enhance safety and quality control throughout the disassembly process. Yang Hu et al. [26] proposed a knowledge recommendation system. This system employed a human–robot collaborative disassembly knowledge graph to assist human workers in disassembly operations and improve disassembly efficiency. The application of knowledge graphs was demonstrated. A review article discussed graph-based disassembly sequence planning [27]. Blocking Graph (with some variations), AND/OR Graph, Liaison Graph, Connector-based Graph methods, and other graph-based methods such as Contact State Graph were outlined.

3. Knowledge Acquisition and Semantic Modelling of EV Batteries

Knowledge graphs can be divided into general knowledge graphs and domain knowledge graphs. A general knowledge graph provides support services for horizontal needs. It emphasizes horizontal coverage and encompasses a wide range of entities from various domains. The relationships between these domains form a multi-dimensional knowledge graph. Domain knowledge graphs provide services for vertical requirements and emphasize deep coverage [28]. Knowledge graph construction relies on both data and ontology models. The ontology model serves as the backbone, defining the conceptual structure. Data acts as a support, providing necessary information. The relationship between ontology, data, and the knowledge graph can be expressed using the following formula:

$$\text{Knowledge Graph} = \text{Data} + \text{Ontology} \quad (1)$$

The formula shows that the knowledge graph is a combination of ontology and data. The knowledge graph is composed of a scale-forming triplet that contains entities and relationships between entities. The triplet can be represented using the following formula:

$$\text{Triplet} = \text{Entity} \overset{\text{Relation}}{\rightleftarrows} \text{Entity} \quad (2)$$

The construction process of the EV battery knowledge graphs in this paper is shown in Figure 1. Firstly, the basic data about EV batteries should be cleaned and preprocessed, and the ontology model needs to be built. The next step is to perform entity extraction and relationship extraction on the processed data. Then, the data are entered into the construction of the graph layer. Finally, the constructed knowledge graph can be applied recommendation systems and sequence planning. A case study was carried out for the robotic disassembly sequence planning of EV batteries using knowledge graph.

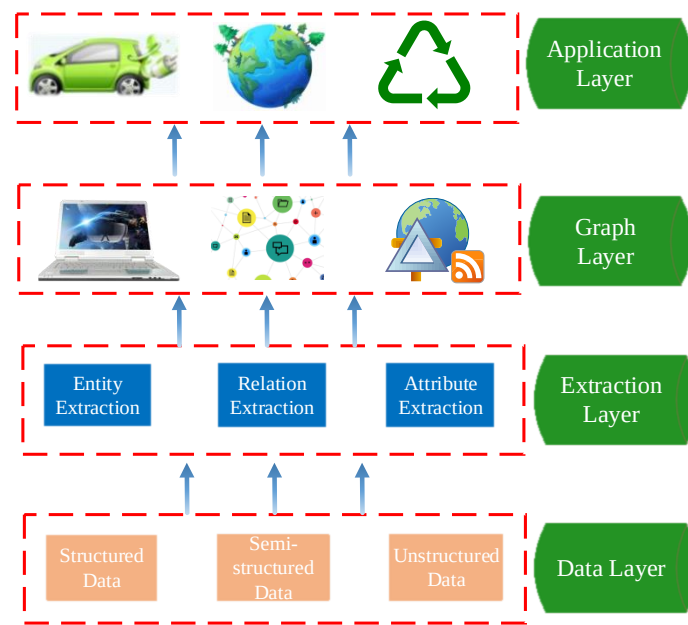


Figure 1. Knowledge graph construction process of EV batteries.

3.1. Semantic Modelling of EV Battery Knowledge

In this paper, based on the structural information of EV batteries, the OWL ontology language is used to construct a structured knowledge description model [29,30]. The structure model describes the basic structure of EV batteries. It takes the overall structure as the first level category, which is divided into six sub-ontologies: lithium battery system, cooling system, connection method, chemical material, electrical components, and sensors, as shown in the following equation:

$$\text{Overall_structure} = \left(\begin{array}{l} \text{lithium_battery_system, cooling_system, connection_methods,} \\ \text{chemical_material, electrical_components, sensors} \end{array} \right) \quad (3)$$

The lithium battery system sub-ontology consists of five sub-categories. It can be expressed using the following formula:

$$\text{lithium_battery_system} = \left(\begin{array}{l} \text{barrier_device, battery_pack_level,} \\ \text{cell_type, packing_device, wiring_harness} \end{array} \right) \quad (4)$$

The cooling system sub-ontology is composed of three sub-categories. This sub-ontology can be expressed using the following equation:

$$\text{cooling_system} = (\text{cooling_methods, design_methods, equipment}) \quad (5)$$

The chemical material sub-ontology consists of three sub-categories, which can be explained using the following formula:

$$\text{chemical_materials} = (\text{battery_materials, connection_materials, fixing_materials}) \quad (6)$$

The connection method sub-ontology is made up of three sub-categories. It can be expressed using the following equation:

$$\text{connection_methods} = (\text{rivet, screwing, weld}) \quad (7)$$

The ontology editing software Protégé 5.5.0 can be utilized to describe and model the semantic model of EV battery knowledge. The specific ontology description model of EV battery knowledge is shown in Figure 2.

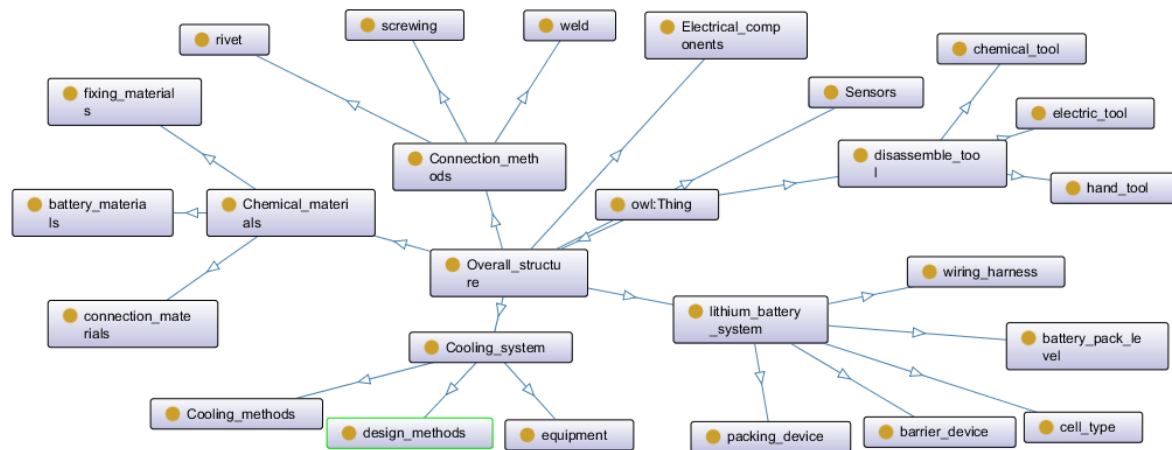


Figure 2. EV battery knowledge ontology description model.

3.2. Knowledge Acquisition of EV Batteries

The data sources in this paper are academic research papers and reports on EV batteries, EV manufacturers, and EV battery suppliers [31]. The data types of battery construction documents include unstructured data, semi-structured data, and structured data. After converting structured data into the JSON format, the data are stored in a CSV file. It is necessary to clean semi-structured data by removing stop words, handling missing values, and removing duplicate words before converting them to a CSV file. The unstructured data require pre-processing, segmentation, followed by entity extraction. For attribute extraction, a portion of the extracted entity serves as attribute data for the node. Knowledge extraction techniques can be utilized to obtain useful information from complex data types. Relationship extraction is the extraction of the relationships between entities from the text. The main constructs of an EV battery are the metal case, battery packs, and wires. Table 1 lists the results of entity extraction, while entities exist as nodes in the knowledge graph.

Table 1. Examples of partial entity nodes.

Entity	Ontology Type	Description of the Entity
Top Fire Guard	Barrier Device	Isolation guards for fire protection on top of the battery
Resistance Welding	Weld	One of the welding methods for joining battery components
Coolant	Chemical Materials	The main components of the battery cooling system
Cylindrical Battery	Cell Type	One of the battery cell types, e.g., 18,650 cells
Sheet Cell	Battery Pack Level	Automotive battery packs can be divided into different-sized cells
Quick Fuse	Electrical Components	A type of fuse that protects short circuits in batteries

The relationship extraction between entities extracts from text data. The entities are connected by various relationships in the knowledge graph which constitute a mesh knowledge system of EV batteries. Table 2 lists some of the results of relationship extraction.

Table 2. Examples of relationships between components of EV batteries.

Relationship	Description of the Relationship
Protection	Covering relationship, e.g., a metal casing protects the entire battery structure.
Connection	Connection relationship, including screw fastening, welding and bonding, etc.
Inclusion	Inclusion relationship, e.g., the battery management system contains several components.
Fixation	Fixation relationship, e.g., the battery module holder fixes the battery module by bonding.
Material	Attribute relationship, e.g., the material composition of a component, e.g., high ignition point material.

in sync with the ontology structure portrayed in Figure 2. Due to the varying ontological structures employed, knowledge graphs are formed with distinct presentation patterns. However, as long as identical data are applied, these two different knowledge graphs can be considered to contain the same content, albeit having contrasting structures. The presentation of the knowledge graph is not limited to a specific structure. Structural changes can be made to the graph under different application scenarios. As shown in Figure 5, this graph is structurally inconsistent with the graph in Figure 3.

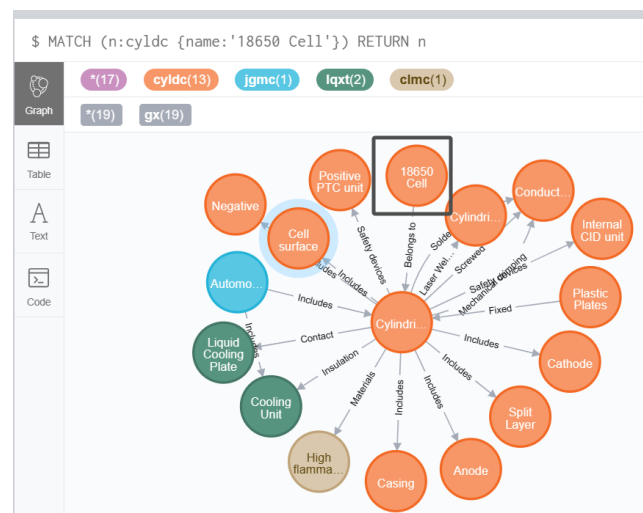


Figure 4. Node “18650Cell” query example.

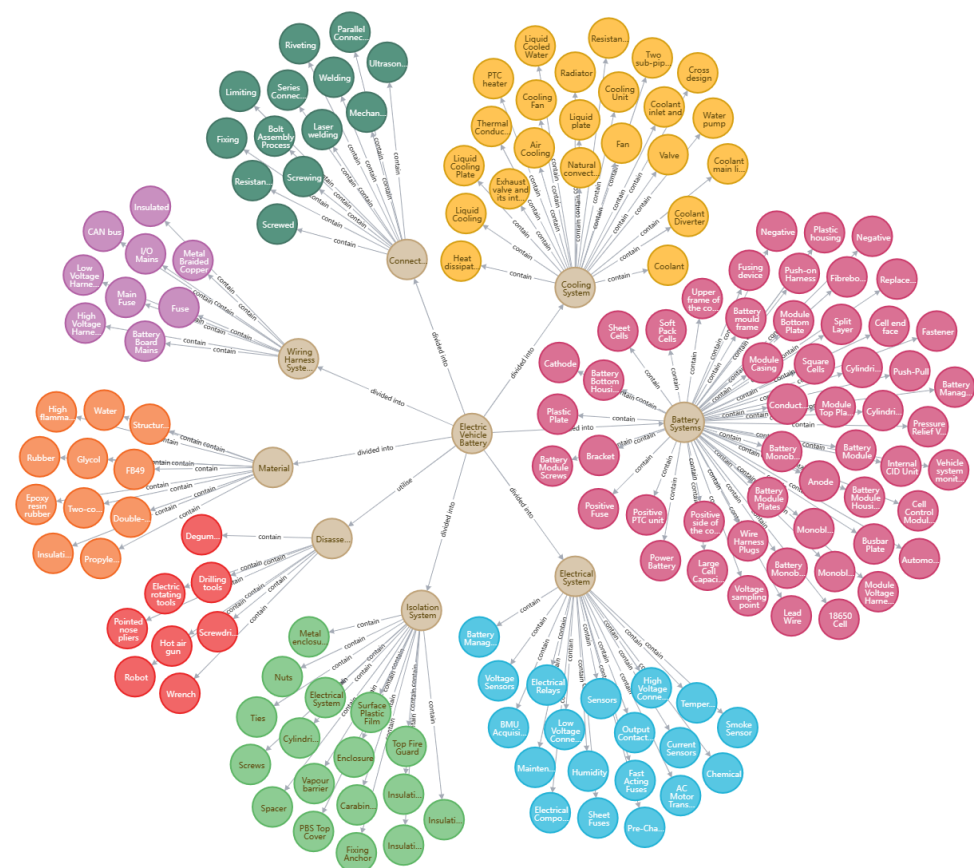


Figure 5. Structure of EV battery knowledge graph.

4. A Case Study of Knowledge Graph Used for Robotic Disassembly of an EV Battery

Knowledge graphs have been widely used in decision support, recommender systems, finance, healthcare, etc. [34,35]. The constructed EV battery knowledge graph can be applied to plan battery disassembly routes. Using the battery knowledge graph, operators can obtain the relationship and connection between various components. It assists operators in determining a reasonable disassembly sequence. The photo of a battery pack from Tesla is shown in Figure 6 to explore the knowledge graph application in disassembly planning. The economic and environmental impacts were considered in the disassembly sequence planning in this section. References were drawn from videos and literature on the disassembly of EoL batteries. The structural information obtained in the previous section and the knowledge graph was combined.

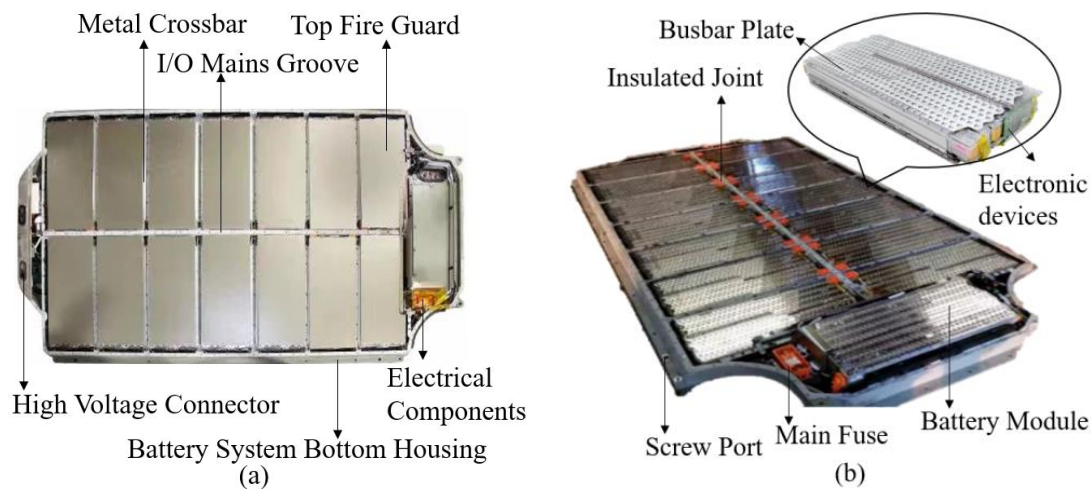


Figure 6. The explored battery pack object. (a) EV battery without the plastic protective film. (b) EV battery after removing the top fire guard.

The following assumptions are made to specify disassembly sequence planning. It is assumed that the parts utilized for disassembly during remanufacturing were in good condition. Before the disassembly process, the batteries were discharged to prevent high-voltage injuries. All processing procedures conformed to the required standards to prevent environmental harm. Parts can be reused, remanufactured, or recycled through physical or chemical treatment, resulting in the majority of materials being reused.

The whole battery system can be divided into five major parts: isolation system, cooling system, wiring harness system, electrical system, and battery system, as listed in Table 3.

Table 3. The main components of the battery pack.

No.	Component Name	Affiliated Systems
1.1	Surface plastic film	Isolation System
1.2	Housing Screws	
1.3	Top Fire Guard	
1.4	Metal housing	
1.5	Electrical system housing	
1.6	Insulation mat	
1.7	Vapor barrier	
1.8	Compartment	

Table 3. *Cont.*

No.	Component Name	Affiliated Systems
2.1	Coolant main line	Cooling System
2.2	Cooling Ducts	
2.3	Coolant	
2.4	Liquid Cooling Boards	
2.5	Thermally Conductive Silicone Sheet	
2.6	Heat Dissipation Copper Tube	
2.7	Exhaust valves and their connections	
3.1	Main Fuse Wire	Wiring Harness System
3.2	I/O Mains	
3.3	Battery Board Main Wiring	
3.4	Braided Copper	
3.5	Fuse Wire	
3.6	High Voltage Harness	
3.7	Low Voltage Wiring Harness	
4.1	Output Contactor	Electrical System
4.2	High Voltage Connectors	
4.3	Low Voltage Connectors	
4.4	Electric Relay	
4.5	BMU Acquisition Boards	
4.6	Sensor	
4.7	BMS Boards	
4.8	Maintenance Switch	
5.1	Battery Module Screws	Battery System
5.2	Battery Module Plate	
5.3	Battery Module	
5.4	Module top guard	
5.5	Module bottom guard	
5.6	Battery Module Housing	
5.7	Busbar Plate	
5.8	Module Voltage Harness	
5.9	Monobloc Split Case	
5.10	Battery Bottom Housing	
5.11	Individual Module Shells	
5.12	Battery Monobloc	

Based on the above internal classification of the EV batteries, components, and the constraint relationships between components, the knowledge graph can be used to determine the nodes corresponding to the component to be disassembled in the next step, until the node corresponding to the last part to be disassembled is determined. Different disassembly routes can be generated for a task, which can be optimized. Figure 7 explains the level of nodes as an example of an EV battery knowledge graph.

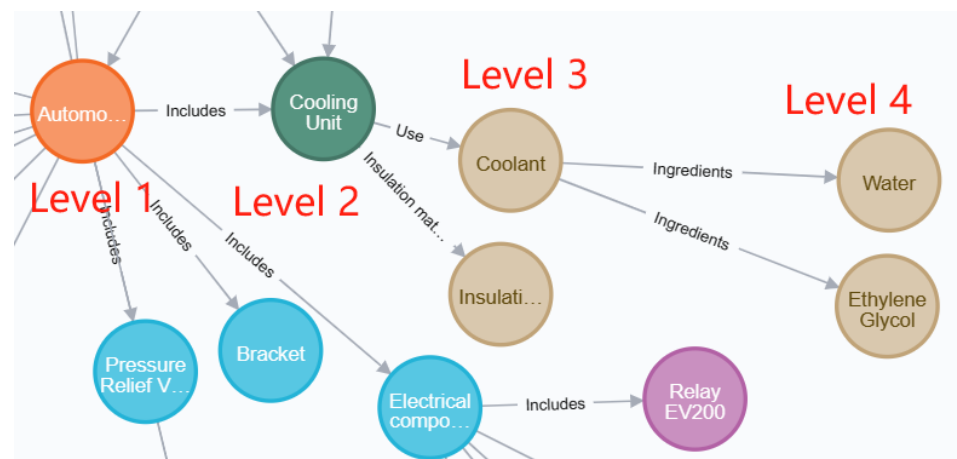


Figure 7. The level of nodes of EV battery knowledge graph.

The disassembly sequence planning of the knowledge graph for EV batteries refers to the step-by-step disassembly and the planning based on internal structure and component characteristics to obtain an efficient, safe, and sustainable battery disassembly process. The disassembly route is shown in Figure 8. Different models of batteries have different structures, so the orders can be changed and some steps can be deleted or added to improve the operation. The route starts at the node “surface plastic film” and ends at the node “battery bottom housing”.

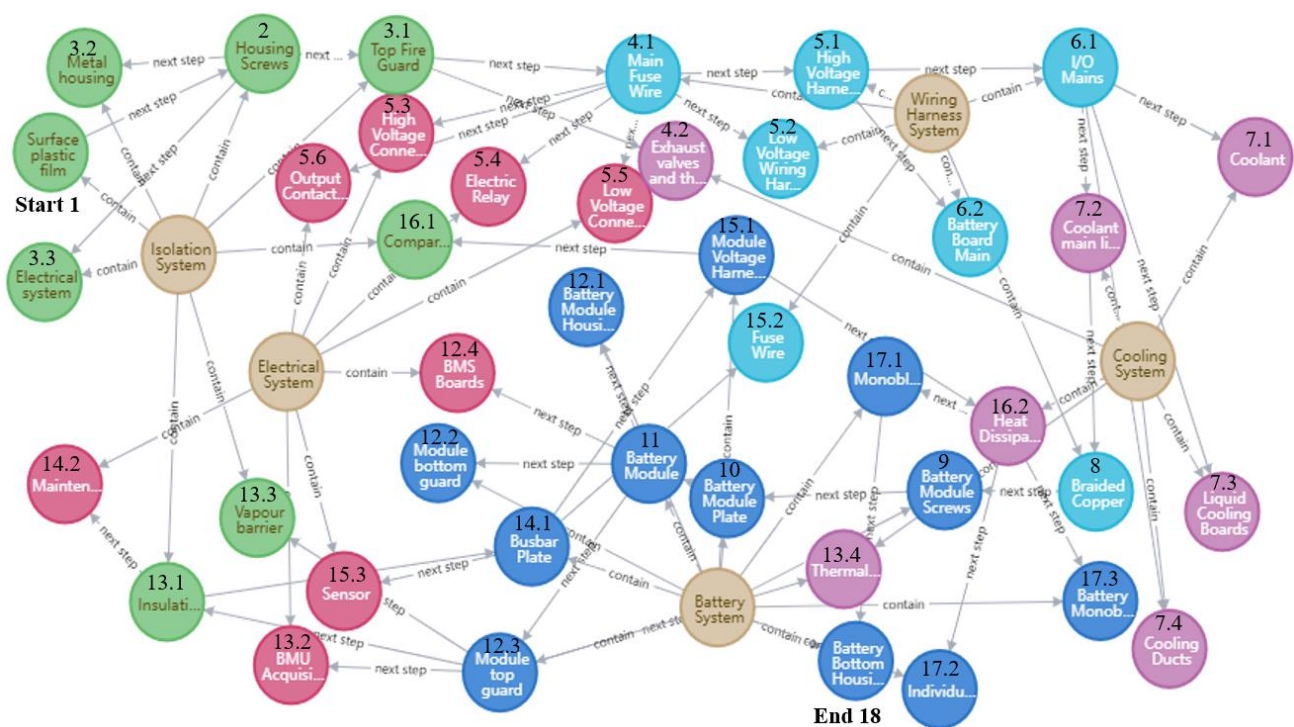


Figure 8. Disassembly sequence planning route of an EV battery.

According to the planned disassembly sequence, the disassembly operations can be described in Table 4:

Table 4. Disassembly sequence.

No.	Specific Operation
1	Peel off the surface plastic film.
2	Remove the housing screws to access the internals.
3	Proceed by disassembling the top fire guard, metal housing, and electrical system housing.
4	Separate the main fuse wire and disconnect the exhaust valves along with their connections.
5	Unplug the high-voltage and low-voltage wiring harnesses, output contactor, high-voltage connector, low-voltage connector, and electric relay.
6	Move on to disconnecting the I/O mains and battery board main wiring.
7	Detach the coolant main line, cooling ducts, and liquid cooling boards.
8	Loosen and remove the braided copper.
9	Unscrew the battery module screws.
10	Detach the battery module plate.
11	Extract the battery module and separate the module top guard, module bottom guard, battery module housing, and BMS boards.
12	Remove the insulation mat, vapour barrier, thermally conductive silicone sheet, and BMU acquisition boards.
13	Disconnect the busbar plate and maintenance switch.
14	Untangle the module voltage harness, fuse wire, and sensor connections.
15	Disassemble the heat-dissipation copper tube and compartment.
16	Access the monobloc split case, individual module shells, and battery monobloc.
17	Ultimately, reach the battery bottom housing.

5. Conclusions

To increase the efficiency and intelligence level of planning robotic disassembly sequences, this paper proposes the method of knowledge graph construction for EV batteries. Firstly, the structure information of the relevant battery was obtained, and the extracted data were combined with the semantic model of ontology structure. Then, the method of knowledge graph construction of EV batteries was introduced. The knowledge graph is a structural data network for representing various concepts, entities, attributes, and relationships of the battery structure in the battery disassembly process. Finally, a case study was carried out to demonstrate the constructed knowledge graph for planning the disassembly route of an EV battery.

The EV battery knowledge graph visualizes the relationships between the parts or components to be disassembled. The resources covered by the knowledge graph can provide operators with quick and accurate queries for relevant information for disassembly knowledge acquisition and decision-making. The proposed knowledge graph could be used to improve the disassembly efficiency of EoL EV batteries.

The disassembly sequence planning using the EV battery knowledge graph includes the following objectives:

1. Enhancing the battery recycling process: Disassembly sequence planning through the knowledge graph can improve the efficiency of battery recycling.
2. Optimizing battery reuse: The EV battery knowledge graph can be applied for the identification of components and materials within batteries that can be further reused to improve battery reuse rates and battery lifespan.
3. Safe handling of battery waste: The effective control and disposal of battery waste can be achieved by planning the disassembly sequence to mitigate risks to the environment and human health by employing appropriate safety measures.
4. Delivering training and guidance: Disassembly sequencing planning can be employed for training and guidance, helping operators to comprehend the inherent intricacies and qualities of batteries and to learn correct operational procedures and disassembly techniques.

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J.W.; supervision, J.H. and R.L.; project administration, J.H. and R.L.; funding acquisition, J.H. and R.L. All authors have read and agreed to the published version of the manuscript.

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